MOD-6 SYNCHRONOUS COUNTER

Digital Electronics — Capstone Design

Objective: Design a 3-bit synchronous counter that cycles through six states (0 to 5) and returns to 0. All three D flip-flops share the same clock.

1. State sequence and transition table

Present state Q2Q1Q0	Decimal	Next state D2D1D0	Decimal
000	0	001	1
001	1	010	2
010	2	011	3
011	3	100	4
100	4	101	5
101	5	000	0
110	6 (unused)	000	0 (recovery)
111	7 (unused)	000	0 (recovery)

2. D flip-flop input equations

Let Q2, Q1, Q0 be present-state outputs. D2, D1, D0 are the inputs to the three D flip-flops.

D2 = (Q0 · Q1 · NOT Q2) + (Q2 · NOT Q0 · NOT Q1)

D1 = NOT Q2 · (Q0 XOR Q1)

D0 = NOT Q0 · (NOT Q1 + NOT Q2)

3. K-map minterms and simplification

Using variable order Q2,Q1,Q0 (Q2 is the most significant bit):

D2 = Σm(3,4); D1 = Σm(1,2); D0 = Σm(0,2,4). The unused states 6 and 7 are assigned next state 0 rather than treated as don't-cares, ensuring recovery.

K-map group interpretation:

D2: isolated minterms m3 and m4 → Q0·Q1·NOT Q2 + Q2·NOT Q0·NOT Q1.

D1: minterms m1 and m2 → NOT Q2·(Q0 XOR Q1).

D0: minterms m0, m2, m4 → NOT Q0·(NOT Q1 + NOT Q2).

4. Circuit implementation

Use three edge-triggered D flip-flops labelled Q2, Q1, Q0. Connect all clock pins to the same clock. Connect the combinational next-state logic outputs D2, D1, D0 to the corresponding D inputs. Feed each Q output back to the next-state logic. Provide a reset that initializes the register to 000.

5. Terminal count and wrap-around

State 101 (decimal 5) is the terminal count. Its next state is 000, so the counter wraps around after 5. States 110 and 111 also recover to 000 on the next active clock edge.

6. Verification

Apply reset, then pulse the common clock. Expected output sequence: 000, 001, 010, 011, 100, 101, 000, repeating. Verify that all three bits change only on the active clock edge.

7. Companion Logisim file

The supplied .circ file is a ROM-based combinational next-state lookup block. Input Q[2:0] selects the next-state value D[2:0]. It is not a complete sequential counter until three D flip-flops are connected in a feedback loop with a common clock and reset.

Submission check: Include this design, the actual flip-flop circuit diagram, K-map work, and a short explanation in the single PDF requested by your course. Open and verify the .circ file in your installed Logisim Evolution version before submitting.